

Quantifying Climate Change Effects on Standard Minimum and Maximum Average Temperature Extremes in Bangladesh: A Machine Learning Regression Analysis from Past to Present

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Abstract. The impact of climate change on temperature extremes is particularly significant in regions like Bangladesh, and it is a pressing global concern. To quantify the effects of climate change on Bangladesh's standard minimum and maximum average temperature extremes, this study uses machine learning techniques on historical records from 1981 to 2010 with current data from 2022. By utilising information from the Statistical Yearbook Bangladesh 2022, which includes temperature readings from 44 different stations, this study offers a thorough evaluation of temperature fluctuations in various geographic areas. Several machine learning models, including Random Forest, Ridge Regression, Bayesian Ridge Regression, K-Nearest Neighbours Regression, and an Ensemble Model, are applied as part of the methodology. The following evaluation metrics are used: R-squared (R^2), Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and Root Mean Squared Logarithmic Error (RMSLE). The study's conclusions highlight notable yearly temperature variations and dynamic patterns of climate change in important places like Khulna, Chattogram, Teknaf, and Dhaka. The outcomes show that, compared to other models, the Ensemble Model performs better across evaluation metrics and is the most successful at predicting temperature extremes. Furthermore, comparing the actual and predicted temperatures for a few selected stations reveals significant deviations that underscore Bangladesh's changing climate dynamics. This research enhances our understanding of the impact of climate change on temperature extremes in Bangladesh. This study helps stakeholders, researchers, and policymakers better understand the difficulties brought on by the region's changing climate patterns.

1 Introduction

Bangladesh is a South Asian nation that deals with a wide range of environmental issues, most of which are made worse by the consequences of climate change. Bangladesh's topographical complexity and geographic location make it an ideal place to investigate how climate change affects weather patterns, especially when it comes to standard minimum and maximum average temperature extremes[4][8].

Bangladesh faces pressing socio-economic challenges, rising temperatures, and ecosystem disruption that necessitate quick adaptation measures[17]. This study employs machine learning to gauge climate change effects on Bangladesh's temperature extremes, revealing evolving trends from historical records to current observations and offering key insights into the nation's dynamic climate. Bangladesh, in South Asia, faces a diverse climate influenced by geography[6,10].

This study is driven by the urgent need to comprehend the evolving climate trends in Bangladesh. Focusing on temperature extremes, it aims to unveil nuanced variations in conventional minimum and maximum average temperatures. Using machine learning regression analysis, the study seeks to provide a comprehensive understanding of how climate change has impacted temperature patterns across the nation, extrapolating insights from historical data to current observations for a more informed perspective.

Analyzing climate-driven temperature shifts in Bangladesh is crucial for sectors like infrastructure, health, agriculture, and disaster management[9,5,18,12]. This research offers concise insights for stakeholders and policymakers to develop adaptable strategies in changing climate conditions.

The research methodology utilized in this study is a systematic strategy that includes machine learning regression models and temperature records from 1981 to 2010 derived from dependable data sources. Temperature extremes are analyzed and predicted in Bangladesh using machine learning regression models, including Random Forest Regression, Ridge Regression, Bayesian Ridge Regression, K-Nearest Neighbors Regression and an Ensemble Model.

This research has ramifications that go beyond Bangladesh. Using machine learning techniques for climate analysis provides a reproducible model for comprehending temperature variations and their effects in other sensitive places worldwide as climate change continues to impact regions internationally. The knowledge gained from this research could help shape climate-related decision-making procedures and support proactive steps to mitigate and adapt to the effects of climate change.

2 Literature Review:

Khan et al. ([14]) conducted a comprehensive study assessing the impact of global warming on climate extremes in Bangladesh, a highly vulnerable country to climate change. Using a subset of the Expert Team on Climate Change Detection and Indices (ETCCDI) index set, the study investigated extreme temperature and precipitation trends at warming levels of 1.5°C, 2°C and 4°C. Based

on high-resolution regional climate model ensembles, their analysis showed significant increases in temperature rise, extreme rainfall, and maximum rainfall, along with upward and downward shifts in warm and cold indices. The study has limitations due to biased data constraints complex and uncertain climate models, but it still predicts more extreme rainfall and extended dry spells in the north-east and southeast. Challenges include downscaling issues, constrained emission scenarios, and limited variable focus, impacting the study's comprehensiveness and practical relevance.

In recent years, the adoption of machine learning (ML) techniques has surged, particularly in research projects exploring the practical applications of wearable technology and algorithms for gait data analysis. Huntingford et al. ([11]) categorize these approaches into wearable, ambient, and vision-based methods. Notably, historical trends in climate change have been revealed through ML regression analyses. Climate change is still a challenging scientific issue despite significant advancements in numerical weather forecasting since the 1950s. They proposed machine learning (ML) and artificial intelligence (AI) as tools for climate research, emphasizing their potential for addressing discrepancies in Earth System Models (ESMs). The parameterization of sub-grid processes in ESMs presents difficulties, prompting researchers to look into ML and AI techniques to reduce inter-ESM uncertainty. Despite the advantages, challenges such as the need for high-quality and diverse datasets, as well as the inherent complexity of climate systems, must be taken into account. The complexity of these systems may make it difficult for ML and AI models to accurately depict real-world situations.

Kafy et al. ([13]) discovered a significant relationship between urban transformations and elevated Land Surface Temperature (LST), posing challenges for environmental engineers and urban planners battling the Surface Urban Heat Island (SUHI) phenomenon. Utilizing the Support Vector Machine (SVM) algorithm and Landsat thermal bands, the study scrutinized Land Use/Land Cover (LULC) variations from 1999 to 2019 due to rapid urbanization. Cellular Automata (CA) machine learning algorithms were used to simulate LULC and seasonal LST scenarios for 2029 and 2039, and the CA model demonstrated excellent accuracy with an overall kappa value of 0.82. MSE and correlation coefficient (R) values of 0.523 and 0.796 in summer and 0.6023 and 0.831 in winter for 2029 and 2039, respectively, were obtained from Artificial Neural Network (ANN) model validation. Simulations predict a notable urban area increase by 9.23% (2029) and 13.59% (2039) compared to 2019, with temperatures exceeding 36°C. This emphasizes the connection between urbanization and rising LST. However, due to potential limitations in satellite data quality and uncertainties in climate models, caution is required.

Mahabub et al. ([15]) (2021) concentrate on the application of machine learning (ML) models for weather forecasting in Bangladesh, a region known for frequent weather changes. Their novel approach entails using ensemble regression algorithms on a raw dataset collected from the Bangladesh Meteorological Division from 2012 to 2018. The dataset includes variables such as wind speed,

humidity, temperature, and rainfall from 33 weather stations. The DTR and Cat-Boost algorithms outperform existing literature metrics using ML-based regression algorithms, according to the study. However, ensemble models for weather prediction face challenges such as a lack of historical data, high computational demands, and uncertainty in representing future climate patterns, which is exacerbated by the dynamic nature of weather systems.

Oyshee et al. ([16]) advocate integrating machine learning with demand-side management (DSM) strategies for modern grid management amidst rising electricity prices. The study finds strong correlations (0.84, 0.87, 0.89) between meteorological data and a synthesized dataset of Bangladesh’s national grid load consumption using the Pearson correlation coefficient. The study finds temperature dependencies and patterns in load consumption using k-Means clustering. Theoretical load shifting potentials are demonstrated by a peak shaving and load shifting algorithm, which produces percentages of 8.83, 9.07, and 8.79 for 2018, 2019, and 2021. Although the paper presents a machine learning model for clustering load and temperature behavior, it also notes limitations in terms of generalization, limited datasets and data quality. It acknowledges possible shortcomings in representing the dynamics of real-world grids, outside influences, interpretability, and viability of widespread adoption. Nevertheless, the research lacks comprehensive validation against alternative models and ignores behavioral and social factors that impact load-shifting strategies’ effectiveness.

In order to mitigate climate change and suggest preventive measures, Shams et al. ([19]) highlight the critical role that weather forecasting plays, particularly in predicting temperatures. The researchers use machine learning techniques to forecast global temperatures based on ten features: aerosols, TSI, MEI, year, month, CH₄, N₂O, CFC.12, and CFC.11. The relationship between the average global temperature and several variables is investigated using a variety of machine learning regressors, such as Linear Regression (LR), Random Forest (RF), Decision Tree (DT), K-Nearest Neighbour (KNN), Support Vector Machine (SVM), and Cat Boost Regressor (CBR). Notably, the CBR outperformed recent machine learning approaches, yielding impressive results. The Cat Boost Regressor achieved Mean Square Error (MSE), Root Mean Square Error (RMSE), Minimum Absolute Error (MAE), and R^2 determination coefficients of 0.003, 0.054, 0.0036, and 92.40%, respectively. Climate models, on the other hand, face challenges such as limited data quality, generalization issues, stationarity assumptions, and difficulty capturing the complexity of climate systems, introducing uncertainties and potentially ethical concerns in prediction misuse and unintended consequences of policy decisions.

3 Methodology

In this section, we outline the methodology utilized in this study. The proposed methodology encompasses several key steps: data collection, data preprocessing and model selection. Figure 1 provides a graphical representation of the proposed methodology.

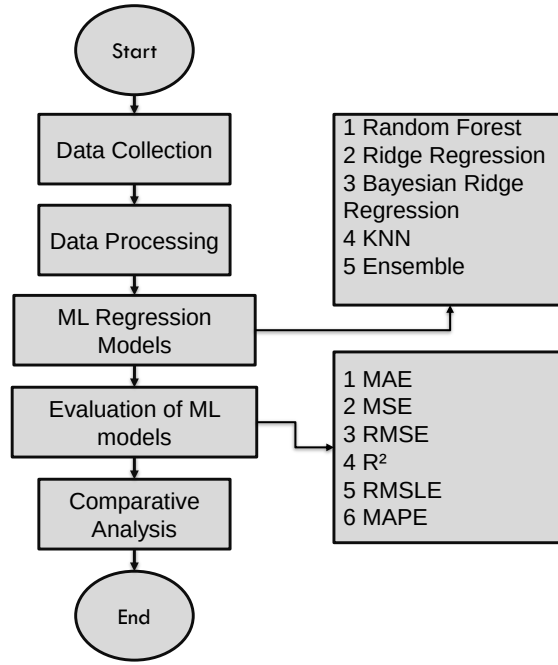


Fig. 1: Overview of the methodology employed in this study for standard minimum and maximum average temperature prediction in Bangladesh.

3.1 Data Collection and Processing

The dataset employed in this study was derived from the Statistical Yearbook Bangladesh 2022, 42nd edition pages 30-34, a comprehensive compilation containing vital information on various aspects of Bangladesh, including land, environment, area, population, agriculture, fishery and more [7]. Specifically, our study focused on the standard minimum and maximum average temperature records for each month spanning 1981-2010. Moreover, an independent dataset was employed for model testing, containing values specifically related to the standard minimum and maximum average temperatures recorded monthly throughout the year 2022. The temperature records are arranged annually and monthly to give a thorough overview. The dataset is noteworthy since it includes 44 distinct stations carefully chosen to reflect regions with larger populations. These stations include prominent places like Dhaka, Chittagong, Cox's Bazar, Rajshahi, Dinajpur, Khulna, Teknaf, and other areas. Including diverse geographical locations ensures a comprehensive representation of temperature variations across Bangladesh. A snapshot of the dataset is provided in Figure 2. After collecting the data, it was checked for null and missing values. The employed dataset contains no null values and no missing values.

1.26 Monthly Average Maximum Temperature in Bangladesh, 2022

(Celsius)

Station	Month											
	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Dhaka-41923	25.1	27.2	33.9	34.2	33.6	33.2	34.1	34.2	33.3	32.7	31.0	27.4
Tangail-41909	23.8	25.9	33.4	34.2	33.1	33.5	34.1	34.2	33.0	32.4	30.6	26.5
Faridpur-41929	24.3	26.7	33.8	34.5	33.4	33.5	34.5	34.0	33.1	32.2	30.5	27.0
Madaripur-41939	23.8	26.4	33.2	34.5	34.0	33.4	33.8	33.5	32.9	31.6	29.6	26.4
Gopalganj-41938	24.1	26.8	33.3	34.6	33.4	33.1	33.8	33.4	33.0	31.7	29.5	26.7
Nikli-41902	24.1	25.7	32.9	33.0	32.0	31.3	33.4	33.3	32.6	31.6	30.1	26.1
Mymensingh-41886	24.1	25.7	32.9	33.0	32.0	31.3	33.4	33.3	32.6	31.6	30.1	26.4
Netrokona-41888	24.2	25.1	32.0	32.1	31.9	31.0	33.5	33.7	32.4	32.0	30.4	26.8
Chattogram-41978	24.6	25.0	32.4	31.5	31.2	30.1	33.2	33.7	32.2	31.8	30.6	28.5
Sandwip-41964	26.0	27.9	32.4	33.1	33.2	32.1	32.8	33.1	32.6	32.8	31.5	28.9
Sitakunda-41965	26.4	28.4	33.4	32.6	32.6	31.5	33.3	33.3	32.5	33.2	32.0	29.2
Rangamati-41966	26.8	28.5	33.8	33.1	32.7	31.7	32.7	33.0	42.9	32.2	28.2	
Cumilla-41933	25.9	25.5	34.5	34.8	32.6	31.8	33.2	33.5	32.9	32.9	31.4	28.0
Chandpur-41941	25.4	27.1	32.8	32.9	33.1	31.6	33.7	33.5	33.0	32.8	31.8	27.4
M Court-41953	25.6	27.7	34.1	34.3	33.9	33.1	33.7	33.8	33.1	33.0	31.3	27.1
Feni-41943	24.7	26.8	33.2	33.8	33.4	32.3	34.0	34.0	32.7	31.9	30.6	28.5
Hatiya-41963	26.0	28.5	34.0	33.6	32.8	31.7	33.4	33.2	32.6	32.8	31.8	27.7
Cox's Bazar-41992	25.2	27.4	33.0	33.3	33.0	31.5	33.6	33.7	31.7	32.0	30.5	29.9
Kutubdia-41989	27.2	28.6	32.7	33.5	32.9	31.2	31.8	31.9	31.9	33.0	32.3	29.0
Teknaf-41998	25.8	28.1	31.7	33.1	32.8	32.0	32.2	31.4	32.3	32.7	31.6	30.3
Sylhet-41891	29.0	30.0	33.3	33.6	32.8	31.7	33.0	32.7	31.5	32.3	31.6	27.3
Srimangal-41915	25.9	26.7	33.5	31.9	31.1	29.7	32.0	31.8	33.3	32.1	31.4	28.0
Rajshahi-41895	25.4	26.9	33.2	34.0	32.4	32.0	34.0	34.2	33.6	32.2	31.7	26.8
Ishurdi-41907	23.4	26.0	34.8	36.2	34.1	35.0	33.7	33.9	34.0	33.2	30.4	26.5
Bogura-41883	23.3	25.7	33.9	35.8	33.7	34.4	36.3	34.8	33.7	32.6	30.3	26.7
Badalgachi-41881	23.5	25.5	33.3	33.4	32.9	32.5	35.0	34.4	32.8	32.7	31.0	26.4
Tarash-41897	22.8	24.8	32.7	33.0	32.6	32.4	34.1	33.8	32.6	31.9	30.1	26.0
Rangpur-41859	23.2	25.3	32.1	33.3	33.1	34.0	34.1	34.4	33.5	31.7	30.6	26.5
Dinajpur-41863	23.5	24.5	32.2	30.6	32.1	31.9	34.3	33.9	32.8	32.0	30.1	26.7
Sayedpur-41858	22.2	24.5	32.4	31.7	32.5	32.2	34.8	34.0	32.8	32.3	30.3	27.3
Tetulia-41850	23.3	25.2	33.0	31.4	33.2	32.3	34.6	33.6	33.6	33.2	30.6	26.5
Dimla-41851	23.2	24.1	32.4	30.5	32.1	30.9	35.6	34.7	32.3	31.2	29.8	26.2
Rajarhat-41856	23.0	24.0	31.4	29.7	31.6	30.6	33.8	33.7	32.3	31.4	29.8	26.2
Khulna-41947	23.5	24.0	32.3	29.0	31.0	30.5	33.7	33.6	32.3	31.2	29.2	27.5
Mongla-41958	24.3	27.3	33.7	34.5	34.9	33.7	33.8	33.5	33.3	32.6	30.7	27.6
Satkhira-41946	24.6	27.9	33.9	35.0	34.5	33.8	33.8	33.6	33.3	32.2	30.2	27.0
Jashore-41936	23.8	26.7	32.9	34.4	34.2	34.0	33.8	33.2	32.9	32.0	30.1	27.9
Chuadanga-41926	24.6	27.6	34.0	35.5	34.8	34.8	32.9	32.8	33.8	32.8	30.9	26.6
Kumarkhali-41927	23.3	26.0	33.8	35.4	34.3	34.7	34.6	33.9	33.8	32.8	30.1	26.5
Barishal-41950	23.3	25.9	34.1	34.8	33.3	33.7	34.6	33.9	33.2	32.4	29.9	27.9
Patuakhali-41960	25.3	27.9	33.5	33.9	33.7	33.1	33.8	33.6	32.9	32.4	30.9	27.9
Khepupara-41984	25.2	28.1	33.7	34.2	33.8	33.0	33.2	32.9	32.8	32.3	30.6	28.2
Bhola-41951	25.7	28.3	33.6	33.7	33.7	32.8	32.8	32.9	32.5	32.4	30.5	28.0

Source: Bangladesh Meteorological Department.

Fig. 2: Snapshot of the dataset employed in this study.

“After collecting the data, we conducted preprocessing on three datasets: the first one covering the years 2017 to 2022, the second related to standard monthly total rainfall, and the third containing historical data from 1981 to 2010. During preprocessing, we addressed null values by removing them. Initially, the Monthly Rainfall dataset had 12 columns, but after preprocessing, it included 13 columns due to the addition of a target column. We performed exploratory data analysis (EDA) to understand the data’s structure, distribution, and characteristics. Additionally, we applied data scaling to the target variables in each dataset. The dataset was split into three categories: training (80

3.2 Machine Learning Model Selection

Random Forest Regression Random Forest Regression is an ensemble learning method that makes accurate predictions by combining the strength of several

decision trees. In our study, the Random Forest model excels at capturing non-linear climate patterns. It aggregates predictions from multiple decision trees, providing a robust and reliable estimation of temperature extremes.

The Random Forest model can be mathematically expressed as follows:

$$\hat{Y} = \frac{1}{n} \sum_{i=1}^N T_i \quad (1)$$

Where $\hat{Y}$ represents the predicted temperature extreme, N is the total number of decision trees in the ensemble, T_i denotes the predicted temperature extreme from the i -th decision tree.

Ridge Regression Ridge Regression, a linear regression modification, addresses variability in climate data by including a penalty term in the loss function, improving our study's model resilience. The model is formulated as follows:

$$\hat{Y} = X\beta + \lambda \sum_{j=1}^p \beta_j^2 \quad (2)$$

Where $\hat{Y}$ represents the predicted temperature extreme, X is the feature matrix, β is the vector of coefficients, λ is the regularization parameter and p is the number of features.

Bayesian Ridge Regression A probabilistic interpretation of predictions is made possible by Bayesian Ridge Regression, which extends linear regression with a Bayesian framework. Its efficacy in handling multicollinearity makes it well-suited for our study on climate data. The model can be expressed as follows:

$$\begin{aligned} \hat{Y} &= X\beta + \epsilon \\ \beta &\sim \mathcal{N}(0, \alpha^{-1}\mathbb{I}) \\ \alpha &\sim \gamma(a, b) \end{aligned} \quad (3)$$

Where $\hat{Y}$ represents the vector of predicted temperature extremes, X is the feature matrix, β is the vector of coefficients, ϵ is the error term, α is the precision parameter, $\mathcal{N}$ denotes a normal distribution, γ represents a gamma distribution and $\mathbb{I}$ is the identity matrix

K-Nearest Neighbors (KNN) Regression A non-parametric, instance-based learning technique called K-Nearest Neighbors (KNN) Regression makes predictions about the future based on how close between two data points are. In the context of our study, The following equation describes KNN Regression:

$$\hat{Y} = \frac{1}{K} \sum_{i=1}^k Y_i \quad (4)$$

Where $\hat{Y}$ represents the predicted temperature extreme, k is the number of nearest neighbors, Y_i denotes the temperature extreme of the i -th nearest neighbor.

Proposed Ensemble Model The Ensemble Model, which combines Bayesian Ridge Regression and Ridge Regression, provides a thorough quantitative assessment of the impact of climate change on temperature extremes in Bangladesh. Leveraging the strengths of both techniques, it provides nuanced analyses from historical to contemporary records. This amalgamation yields a robust, flexible framework offering valuable insights into evolving climate patterns. The model can be expressed as follows:

$$\hat{Y}_{Ensemble} = \omega \cdot \hat{Y}_{Ridge} + (1 - \omega) \cdot \hat{Y}_{Bayesian} \quad (5)$$

Where $\hat{Y}_{Ensemble}$ represents the predicted temperature extreme by the Ensemble Model, $\hat{Y}_{Ridge}$ is the prediction from the Ridge Regression model, $\hat{Y}_{Bayesian}$ is the prediction from the Bayesian Ridge Regression model, ω is the weight assigned to the Ridge Regression model, with $0 \leq \omega \leq 1$.

3.3 Hyper-parameter Tuning

Upon completing the data processing phase, selecting a suitable machine learning model for temperature prediction relies on analysing previously recorded data. Each candidate model undergoes testing with the selected features to identify the optimal model. The machine learning models used in this study are shown in Table 1 with the associated hyperparameter settings.

Table 1 summarizes the hyper-parameter configurations for this study's five distinct machine learning models. The Random Forest Regressor is characterized by 150 estimators, a maximum depth of 30, a minimum sample split of 2 and a minimum sample leaf of 1. The Ridge Regressor is defined with a default alpha value of 1.0. For the Bayesian Ridge Regressor, the hyperparameters include alpha_1, alpha_2, lambda_1, lambda_2, fit_intercept and verbose. "fit_intercept" (defaulted to True) determines intercept calculation, while "verbose" (set to True) yields detailed output during model fitting for enhanced training insights. The KNN Regressor utilizes 7 neighbours, 'distance' weights, 'auto' algorithm, a leaf size of 30 and a power parameter of 2 for the Minkowski metric. Lastly, the Ensemble Model incorporates alpha, lambda, compute_score, normalize, verbose and fit_intercept parameters. Where "compute_score" (True computes score during fitting) and "normalize" (True enables input feature normalization) are in the Ensemble Model. This extensive collection of hyper-parameter settings is essential for customizing the behavior of each model to the particular needs of the study's climate change analysis.

3.4 Evaluation Matrices

Following the identification of optimal models for this study, the evaluation of their performance is conducted using six metrics: Mean Absolute Error (MAE),

Table 1: Hyper parameter settings for various machine learning models utilized in this study.

Model Name	Hyperparameter Settings
Random Forest	N-estimators = 150 max-depth = 30 min-samples-split = 2 min-samples-leaf = 1
Ridge Regression	alpha = 1.0
Bayesian Ridge Regression	alpha 1=1e-6 alpha 2=1e-6 lambda 1=1e-6 lambda 2=1e-6 fit_intercept=True verbose=False
KNN Regression	n-neighbors=7 weight:distance algorithm:auto leaf-size=30 p=2
Ensemble Model	alpha=1.0 Lambda=1.0 compute_score=False Normalize=False verbose=False Fit_intercept=True

Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-squared (R²), Root Mean Squared Logarithmic Error (RMSLE), and Mean Absolute Percentage Error (MAPE). The equations for these evaluation metrics are detailed below in Equation 6.

$$\begin{aligned}
MAE &= \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \\
MSE &= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\
RMSE &= \sqrt{MSE} \\
R^2 &= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \\
RMSLE &= \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\log(1 + y_i) - \log(1 + \hat{y}_i) \right)^2} \\
MAPE &= \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100
\end{aligned} \tag{6}$$

Where, N is the number of samples, y_i is the actual values, $\hat{y}_i$ is the predicted value and $\bar{y}_i$ is the mean of the actual values.

3.5 Experimental Setup

In this study, the main tool for our machine learning model building, implementation and training was the cloud-based platform Google Colab [2]. The utilization of Google Colab's free version afforded us a robust and readily accessible computing environment, thereby enabling the execution of extensive experiments without necessitating substantial computational resources. The research team's ability to easily share code and datasets was made possible by Google Colab's collaborative capabilities, which promoted effective teamwork and increased overall research efficiency. Notably, all code was scripted in Python3 and the machine learning library employed was Scikit Learn [1,3].

4 Result and Discussion:

This study utilized diverse modeling approaches, employing various combinations, to explore predictive modeling of standard minimum and maximum average temperatures. The investigation focused on a case study conducted in Bangladesh.

Within Table 2 comprehensive evaluation of temperature prediction models using metrics like MAE, MSE, RMSE, R2, RMSLE and MAPE, our ensemble model consistently emerges as the superior performer. In MAE, Bayesian Ridge and Ridge Regression score 0.11, while our ensemble model achieves 0.09. For MSE, our ensemble model excels with 0.01, outshining KNN Regression's 7.03. The RMSE showcases the superiority of our ensemble model with a value of 0.12.

Table 2: Evaluation Metrics for Standard Minimum and Maximum Average Temperature Prediction (MAE, MSE, RMSE, R2, RMSLE, MAPE).

Model Name	MAE	MSE	RMSE	R	RMSLE	MAPE
Random Forest	0.20	0.14	0.37	0.99	0.01	0.88
Bayesian Ridge	0.11	0.02	0.14	1.00	0.00	0.45
KNN Regression	0.88	7.03	2.65	0.65	0.03	4.18
Ridge Regression	0.11	0.02	0.15	1.00	0.00	0.48
Ensemble Model	0.09	0.01	0.12	1.00	0.00	0.37

In R2, KNN Regression and Random Forest score 0.65 and 0.99, respectively, while Ridge Regression, Bayesian Ridge Regression, and our ensemble model all achieve a perfect 1.00. The RMSLE demonstrates the ensemble model's exceptional performance with a score of 0.00. Closing with MAPE, our ensemble model stands out with a score of 0.37, reaffirming its superior accuracy. In summary, the ensemble model consistently outperforms across all metrics, making it the best option for predicting standard minimum and maximum average temperatures.

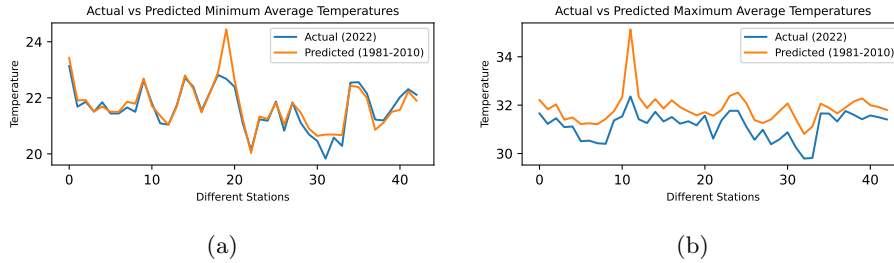


Fig. 3: Annual Comparative Analysis of Actual and Predicted Standard Minimum and Maximum Average Temperatures for 7 Selected Stations Among 44 Locations.

In Figure 3(a), the representation of standard minimum average temperatures across 44 stations reveals noticeable fluctuations compared to historical records. Similarly, Figure 3(b) displays standard maximum average temperatures for the same 44 stations, highlighting discernible variations from records.

Table 3 provides a comprehensive overview of annual temperature variations in Bangladesh, emphasizing key stations. Seven key stations has been chosen based on their dense population. Notably, in Dhaka, the observed minimum average temperature of 23.13 differs from the anticipated value of 23.42, and the maximum average temperature of 31.65 contrasts with the expected 32.21. Similar discrepancies are evident in Teknaf, where the observed minimum average temperature of 22.67 deviates from the projected 24.44, indicating a significant disparity. Additionally, in Khulna, the recorded maximum average temperature

Table 3: Annual Comparative Analysis of Actual and Predicted Standard Minimum and Maximum Average Temperatures for 7 Selected Stations Among 44 Locations.

Section	Min Avg Temo(Actual)	Min Avg Temo(Predicted)	Max Avg Temo(Actual)	Max Avg Temo(Predicted)
Dhaka	23.13	23.42	31.65	32.21
Chattogram	21.50	21.79	30.40	31.43
Cox's Bazar	22.18	22.16	31.23	31.92
Sylhet	22.39	22.63	31.56	31.7
Rajshahi	20.13	20.03	31.36	31.80
Teknaf	22.67	24.44	31.16	31.58
Khulna	20.28	20.67	29.81	31.12

of 29.81 falls short of the projected 31.12. These variations underscore the annual temperature fluctuations relative to historical data, contributing to discernible shifts in climate patterns. The observed intensification of heat during summer and increased cold during winter aligns with broader climate change trends, where temperature serves as a pivotal factor. Results achieved from this study confirms the role of annual standard minimum and maximum average temperatures in shaping Bangladesh’s evolving climate dynamics.

Our study, which extends across a broader time span and incorporates data from more stations compared to a previous analysis [15], highlights the superiority of the proposed ensemble model over baseline machine learning (ML) models. While the previous study achieved moderate accuracy using the CatBoost model, our ensemble model significantly surpassed it in performance, as indicated by substantially lower MAE, MSE, and MAPE values. Our proposed model outperform baseline ML models due to the collective intelligence of multiple models, each capturing different aspects of the underlying data patterns. By aggregating predictions from multiple models, proposed ensemble method mitigate individual model biases and errors, resulting in more accurate and robust predictions. Additionally, the proposed ensemble model offers increased stability and generalization capability, as ensemble models are less susceptible to overfitting compared to single ML models. This advantage of ensemble modeling, coupled with our study’s comprehensive data coverage and longer time frame, underscores the effectiveness of ensemble techniques in improving predictive accuracy and providing deeper insights into climate change dynamics in Bangladesh.

5 Conclusion

This study utilizes a robust machine learning regression analysis to quantify the impact of climate change on standard minimum and maximum average temperature extremes in Bangladesh from past to present. A comprehensive assessment across 44 stations reveals significant annual temperature variations, evidenced by deviations in crucial locations such as Dhaka, Chattogram, Teknaf, and Khulna. These observed discrepancies underscore the region’s dynamic nature of climate change, exhibiting notable shifts in both summer heat and winter cold.

Our findings emphasize the pivotal role of temperature fluctuations in contributing to broader climate change patterns. This study enhances our understanding of Bangladesh's changing climate dynamics, offering valuable insights for policymakers and researchers to address the challenges posed by climate change in the region. Limitations of the study include potential issues with data quality and availability, as MLRA relies on accurate and up-to-date climate data, which may be subject to errors or gaps. Furthermore, the assumption of stationarity in MLRA may introduce bias due to the non-stationary nature of climate systems, where statistical properties can change over time. These results highlight how critical it is to provide focused mental health interventions and assistance to university students in Bangladesh so they can manage the psychological effects of the current pandemic.

Data and Code Availability

The codes and data employed in this study can be found in the following GitHub repository: https://github.com/ShahriarAyon63/st_min_and_st_max_ave_temperature_predic_Bangladesh

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